

Spin Dynamics Modeling Reference

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1 Notation and Conventions

Unless a section explicitly states otherwise, vectors are bold, unit vectors carry a hat, and time is measured in seconds. Magnetic field strength $\mathbf{H}$ is in A m⁻¹, and effective flux density $\mathbf{B}_{\text{eff}}$ is in tesla, with $\mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}$ when they describe the same effective interaction. The following symbols are used throughout the library.

$\mathbf{m}$ Normalized magnetization, with $\|\mathbf{m}\| = 1$.

$\mathbf{s}_i$ Unit spin direction at site i .

M_s Saturation magnetization of the ferromagnetic layer.

$\mathbf{H}_{\text{eff}}$ Effective magnetic field.

σ Spin-polarization unit vector.

α Dimensionless Gilbert damping parameter.

γ Positive gyromagnetic ratio, $\gamma = g\mu_B/\hbar$, where g is the Landé g -factor.

γ_0 Positive gyromagnetic factor for an $\mathbf{H}$ -field formulation, $\gamma_0 = \mu_0 |\gamma|$.

θ_{SH} Spin-Hall angle.

J_c Charge-current density.

t_{FM} Ferromagnetic-layer thickness.

β Signed torque ratio, $\beta = \tau_{\text{FLT}}/\tau_{\text{DLT}}$.

$\tau_{\text{DLT}}, \tau_{\text{FLT}}$ Damping-like and field-like torque amplitudes, respectively; both are in tesla in the LLGS equations.

e Positive elementary charge.

μ_0 Vacuum permeability.

μ_B Bohr magneton.

$\hbar$ Reduced Planck constant.

k_B Boltzmann constant.

T Absolute temperature.

Physical constants. The following values reproduce `constantfile.m`:

$$\begin{aligned} \gamma &= 1.760859644 \times 10^{11} \text{ rad s}^{-1} \text{ T}^{-1}, & e &= 1.602176565 \times 10^{-19} \text{ C}, \\ \mu_0 &= 1.25663706143592 \times 10^{-6} \text{ T m A}^{-1}, & \mu_B &= 9.274 \times 10^{-24} \text{ J T}^{-1}, \\ \hbar &= 6.58211951440 \times 10^{-16} \text{ eV s}, & k_B &= 1.38064852 \times 10^{-23} \text{ J K}^{-1}. \end{aligned}$$

Convention warning. The macrospin effective-field section of *Spin Dynamics Modeling Reference* defines both $\mathbf{H}_{\text{eff}}$ in A m^{-1} and $\mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}$ in tesla. The atomistic Hamiltonian and LLGS sections use $\mathbf{B}_{\text{eff}}$ in tesla with γ in $\text{rad s}^{-1} \text{ T}^{-1}$, as do the macrospin and ASTRO implementations [33, 11]. Convert an $\mathbf{H}$ -field in A m^{-1} to its tesla-valued equivalent before using it in this LLGS equation; a code variable named `Hext` does not determine its units.

2 Representative Material Parameter Sets

The values below reproduce particular numerical models from cited papers and repository implementations. They are model-specific parameter sets rather than universal material constants. In particular, quantities associated with a heavy-metal layer, spin polarization, device geometry, or numerical integration are identified separately from intrinsic magnetic parameters. Entries that do not yet have a source within the approved project reference set are retained provisionally and carry a visible reference warning.

2.1 Property of Heavy Metal

The spin-Hall angle belongs to the heavy-metal spin-current source and is not specific to a ferromagnetic, ferrimagnetic, or antiferromagnetic target layer. For the signed representative values in Table 1, we use $\mathbf{J}_s = \theta_{\text{SH}} \boldsymbol{\sigma} \times \mathbf{J}_c$ and choose Pt as the positive reference; Ta and W then have negative signs [5, 6, 7]. Reversing the current, layer-normal, or spin-polarization convention requires a corresponding sign change in the torque implementation.

The compact-model README compares reported resistivities of $190 \mu\Omega \text{ cm}$ for Ta and $170\text{--}260 \mu\Omega \text{ cm}$ for W, then uses the representative value $\rho_{\text{HM}} = 200 \mu\Omega \text{ cm}$ [5].

Property	Value	Evidence and qualification
Pt spin-Hall angle	+0.07	Representative literature value quoted by Liu et al. and Pai et al.; defines the positive reference sign
High-resistivity β -Ta spin-Hall angle	-0.12 to -0.15	Consistent magnitudes from ST-FMR, perpendicular switching, and three-terminal switching
High-resistivity β -W spin-Hall angle	-0.30 ± 0.02 (ST-FMR); -0.33 ± 0.06 (switching)	Negative sign verified relative to Pt; the ST-FMR value is a lower bound if spin diffusion suppresses the measured torque
Heavy-metal resistivity	$\rho_{\text{HM}} = 200 \times 10^{-8} \Omega \text{ m}$	Representative Ta/W value from the compact model; $\rho_{\text{HM}} = 200 \mu\Omega \text{ cm}$

Table 1: Representative properties of Pt, β -Ta, and β -W heavy metals [5, 6, 7].

2.2 Ferromagnet: CoFeB

The CoFeB free layer is represented by a one-sublattice macrospin LLGS model with perpendicular anisotropy. The anisotropy field and saturation magnetization below are the defaults of the SOT/STT compact model; the damping constant comes from the model of Zhang et al. [5, 4].

Quantity	Value	Role or convention
Uniaxial anisotropy field	$B_k = 1.5 \text{ T}$	Compact-model parameter $\mathbb{H}\mathbf{k}$; $H_k \approx 1.19 \times 10^6 \text{ A m}^{-1}$
Saturation magnetization	$M_s = 1000 \text{ emu cm}^{-3}$	$M_s = 1.00 \times 10^6 \text{ A m}^{-1}$; $\mu_0 M_s \approx 1.26 \text{ T}$
Gilbert damping	$\alpha = 0.02$	Magnetic model parameter

Table 2: Selected CoFeB macrospin parameters from the SOT/STT compact model and Zhang et al. [5, 4].

Parameter justification. Based on Nat. Mater. **9**, 721 (2010)., $H_{\text{an}} = 2(K_{\text{bulk}} + K_i/t_{\text{FL}})/M_s$ with $K_{\text{bulk}} = 2.245 \times 10^5 \text{ J m}^{-3}$, $K_i = 1.286 \times 10^{-3} \text{ J m}^{-2}$, $M_s = 1.58 \text{ T} = 1257 \text{ emu cm}^{-3}$, when we use $t_{\text{FL}} = 1.2 \text{ nm}$, it gives $H_{\text{an}} = 2.06 \text{ T}$. Here we approximate $M_s = 1000 \text{ emu cm}^{-3}$, $H_k = 1.5 \text{ T}$ [5].

2.3 Ferrimagnet: GdFeCo

Zhang et al. use a random two-dimensional atomistic $(\text{FeCo})_{1-x}\text{Gd}_x$ lattice driven by damping-like STT, whereas Zhu et al. use a one-dimensional atomistic GdFeCo model driven by SOT [14, 15]. Both are fixed-length LLG or LLGS models. The 2022 paper states that its parameters are otherwise the same as those in the 2020 paper, but replaces the single exchange coefficient with three bond-specific values. Table 3 therefore keeps the two published exchange parameterizations separately rather than selecting or averaging them.

The exchange values are deliberately not converted into a single common coefficient. The 2020 value belongs to a one-dimensional nearest-neighbor Hamiltonian, while the 2022 values distinguish the chemical identity of each bond in a random two-dimensional alloy. Their lattice topology and bond statistics must be retained with the numerical values when either set is used.

2.4 Noncollinear Antiferromagnet: Mn_3Sn

The three-sublattice macrospin switching model of Xu et al. and the later spatially resolved atomistic domain-wall model use the same magnetic moment, DMI, anisotropy, damping, and

Quantity	Zhu et al. (2020)	Zhang et al. (2022)
Spatial model	1D atomistic lattice with lattice constant $d = 0.4$ nm; strip length and width are 400 and 40 nm	Random 2D square lattice; each interior site has four nearest neighbors; the main phase diagrams use 100 atoms
Exchange	$A_{\text{ex}} = +3$ meV = $+4.8065 \times 10^{-22}$ J	$A_{\text{Gd-Gd}} = -1.26 \times 10^{-21}$ J; $A_{\text{FeCo-FeCo}} = -2.83 \times 10^{-21}$ J; $A_{\text{FeCo-Gd}} = +1.09 \times 10^{-21}$ J
Perpendicular anisotropy	$K_{\text{TM}} = K_{\text{RE}} = 0.04$ meV	same as Zhu et al. (2020)
Hard-axis anisotropy	$\kappa_{\text{TM}} = \kappa_{\text{RE}} = 0.2$ μeV	Not included in the stated Hamiltonian
DMI	$D = 0.128$ meV	Not included in the stated Hamiltonian
Gilbert damping	$\alpha_{\text{TM}} = \alpha_{\text{RE}} = 0.015$	same as Zhu et al. (2020)
Landé factors	$g_{\text{TM}} = 2.2$; $g_{\text{RE}} = 2.0$	same as Zhu et al. (2020)
Sublattice magnetizations	$M_{s,\text{TM}} = 1149$ kA m $^{-1}$; $M_{s,\text{RE}} = 1012$ kA m $^{-1}$ at $T = 300$ K	same as Zhu et al. (2020)
Ferrimagnet thickness	$t_{\text{FiM}} = 0.4$ nm	same as Zhu et al. (2020)
Drive parameter	Spin-Hall angle $\theta_{\text{SH}} = 0.2$; field-like to damping-like ratio β is varied rather than fixed in the baseline list	STT efficiency η is introduced but no numerical value is stated
Compensation composition	Not stated as a composition in the parameter block	$x_{\text{MC}} = 0.23$; $x_{\text{AMC}} = 0.21$

Table 3: Atomistic GdFeCo parameter sets retained in their native model conventions [15, 14].

spin-Hall angle, but different exchange coefficients [8, 10]. The values must therefore remain attached to their model and interaction convention. Both Hamiltonians take positive A to favor antiparallel alignment, and the crystalline easy axes point toward the nearest Sn atoms. The atomistic domain-wall paper additionally states that its positive D favors clockwise order.

Quantity		Value	Model scope or convention
Mn moment		$m_s = 3\mu_B$	Per Mn site in both models
Nearest-neighbor exchange	ex-	$A = 17.53$ meV	Three-sublattice macrospin switching model
Nearest-neighbor exchange	ex-	$A = 9$ meV	Spatial atomistic domain-wall model
DMI coefficient		$D = 0.833$ meV	Same-layer interaction in both models
Crystalline anisotropy		$K = 0.196$ meV	Single-site Hamiltonian coefficient in both models
Gilbert damping		$\alpha = 0.003$	Same in both fixed-length models
Unit-cell volume		$V_{\text{cell}} = 0.3778$ nm ³	Macrospin model; $M_s = 6m_s/V_{\text{cell}} \approx 4.42 \times 10^5$ A m ⁻¹
Mn ₃ Sn thickness		$d = 30$ nm	Macrospin SOT model geometry
Time step		$\Delta t = 5$ fs	Fourth-order Runge–Kutta macrospin integration
Nanoribbon length		5.7 μm	Spatial atomistic domain-wall model geometry

Table 4: Mn₃Sn parameter sets used in the macrospin switching and atomistic domain-wall models [8, 10].

The Mn₃Sn entries are not by themselves a complete crystallographic simulation specification: a reproducible atomistic calculation must also retain the source model’s lattice, site mapping, bond list, DMI-vector orientations, boundary conditions, and pair-counting convention.

2.5 Layered van der Waals Antiferromagnets

The parameter sets in this subsection describe A-type van der Waals antiferromagnets at the layer-macrospin or layer-micromagnetic level. Each ferromagnetic layer is represented by a normalized magnetization, and neighboring layers are coupled antiferromagnetically. The areal interlayer exchange $J_{\perp}$ must not be substituted directly for an atomistic bond energy or a volume exchange stiffness. Values from different source rows are kept separate because they were obtained at different temperatures or with different experimental and computational conventions.

Reference needed. The CrPS₄, MnBi₂Te₄, and CrI₃ exchange-length entries below are provisionally retained from the direct literature compilation in Wang et al. [18]; that paper is not in the project’s currently approved source set. The source-reference number printed by Wang et al. is preserved for every row so that distinct parameter determinations are not averaged or conflated.

2.5.1 CrPS₄ micromagnetic set

Reference needed. The following values reproduce the CrPS₄ micromagnetic and energy-landscape calculations of Wang et al. [18], which is not in the approved project reference set.

Material and source	Condition	d (nm)	$J_{\perp}$ (J m^{-2})	K (J m^{-3})	l_{ex} (nm); N
CrPS ₄ , ref. 30	Experiment, 22.5 K	0.61	1.58×10^{-4}	6.33×10^3	2.77; 4.5
CrPS ₄ , ref. 31	Experiment, 2 K	0.61	2.94×10^{-4}	1.74×10^4	2.27; 3.6
CrPS ₄ , ref. 16	Calculation	0.61	2.75×10^{-4}	1.26×10^4	2.58; 4.2
MnBi ₂ Te ₄ , ref. 44	Experiment, 2 K	1.36	1.90×10^{-2}	1.47×10^6	2.98; 2.2
MnBi ₂ Te ₄ , ref. 45	Experiment, 1.8 K	1.36	6.42×10^{-3}	1.45×10^6	1.73; 1.3
CrI ₃ , ref. 46	Calculation	0.66	1.96×10^{-4}	4.04×10^5	0.40; 0.6
CrI ₃ , ref. 47	Calculation	0.66	3.09×10^{-4}	4.62×10^5	0.47; 0.7

Table 5: Source-specific interlayer exchange and anisotropy sets compiled by Wang et al. [18]. Here d is the monolayer thickness, l_{ex} is the reported interlayer exchange length, and $N = l_{\text{ex}}/d$ is the number of layers within that length. Rows are matched sets and must not be mixed.

Quantity	Value	Scope or convention
Monoclinic anisotropy constants	$K_1 = 12,485 \text{ J m}^{-3}$; $K_2 = 8,897 \text{ J m}^{-3}$; $K_m = 1,271 \text{ J m}^{-3}$	Layer-resolved COMSOL micromagnetic model; corresponding $\mu_0 H_{K1} = 0.167 \text{ T}$, $\mu_0 H_{K2} = 0.119 \text{ T}$, and $\mu_0 H_{Km} = 0.017 \text{ T}$
Saturation magnetization	$M_s = 74,803 \text{ A m}^{-1}$	$\mu_0 M_s = 0.094 \text{ T}$
Interlayer exchange	$J_{\perp} = 1.58 \times 10^{-4} \text{ J m}^{-2}$	Antiferromagnetic; reported equivalent coupling field $\mu_0 H_C = 3.45 \text{ T}$
Intralayer exchange	$J_{\parallel} = 1.48 \times 10^{-2} \text{ J m}^{-2}$	Areal interaction used in the COMSOL model
Gilbert damping	$\alpha = 0.1$	Fixed-length LLG simulation input
Intralayer spin stiffness	$A = 1.5 \times 10^{-12} \text{ J m}^{-1}$	Separate domain-wall energy-landscape calculation
Uniaxial anisotropy	$K = 6.0 \times 10^3 \text{ J m}^{-3}$	Separate domain-wall energy-landscape calculation; do not substitute for the three monoclinic constants above

Table 6: CrPS₄ parameters retained in the distinct model conventions used by Wang et al. [18].

2.5.2 Bilayer CrI₃ staggered-SOT set

Reference needed. The bilayer CrI₃ values below are provisional because Xue and Haney [19] is not in the approved project reference set. This is a two-sublattice, fixed-length macrospin calculation for electron-doped antiferromagnetic bilayer CrI₃, not an experimental device parameterization.

Quantity	Value	Scope or convention
Easy-axis anisotropy field	$H_A \approx 1.77$ T	Source uses a magnetic-field-in-tesla convention
Antiferromagnetic exchange field	$H_E \approx 0.76$ T	Comparable to H_A in the source model
Gilbert damping	$\alpha \approx 0.04$	Common to both fixed-length layer macrospins
Even torkance, $\hat{N} \parallel \hat{z}$	$\sim 1 ea_0/\hbar$	At about 0.1 eV above the conduction-band minimum
Even torkance, $\hat{N} \parallel \hat{x}$	$\sim 0.4 ea_0/\hbar$	Same first-principles torque convention
Switching threshold	$ E \approx 2$ V μm^{-1}	Predicted from the interpolated angular torkance
Representative reversal	$E = -2.3$ V μm^{-1} ; < 100 ps	Damping-like torque drives reversal; field-like torque assists
Oscillatory state	$E = 3$ V μm^{-1} ; ~ 80 GHz	Requires both damping-like and field-like torque components

Table 7: Bilayer CrI₃ macrospin and first-principles SOT parameters from Xue and Haney [19].

2.5.3 (Fe, Co)₃GaTe₂

Guo et al. model each ferromagnetic FCGT layer as one fixed-length macrospin and demonstrate SOT switching in odd-layer Pt/FCGT devices. Their main article reports a normalized exchange field but not independent absolute values of H_{an} and H_{ex} ; only the published ratio is therefore listed below [17].

Quantity	Value	Scope or convention
Néel temperature	$T_N = 196$ K	Bulk FCGT characterization
Exchange-to-anisotropy field ratio	$H_{\text{ex}} = H_{\text{an}}/3$	Experimental-data-derived macrospin input; absolute fields are not stated in the main article
Applied in-plane field in model	$H_{\text{ext}} = 0.4H_{\text{an}}$	SOT-switching simulation input
Gyromagnetic ratio	$\gamma = 1.76 \times 10^{11}$ T ⁻¹ s ⁻¹	Fixed-length Gilbert-form LLG model
Gilbert damping	$\alpha = 0.1$	Selected simulation input, not a reported FCGT measurement
Critical current density	$J_c = 11.4$ MA cm ⁻²	11-layer Pt/FCGT device at 180 K
SOT efficiency	~ 38 Oe (MA cm ⁻²) ⁻¹	Estimated from current-induced hysteresis-loop shift
Spin-mixing conductance	1.93×10^{15} Ω^{-1} m ⁻²	Pt/FCGT interface inferred from spin-Hall magnetoresistance
Field-free switching ratio	$\sim 70\%$ at 120 K	Pt/FCGT/CrSBr heterostructure; partial switching is attributed to a multidomain CrSBr layer

Table 8: FCGT macrospin inputs and experimental SOT figures of merit from Guo et al. [17].

The selected sources provide no damping constant, torque coefficient, or complete LLG parameterization for MnBi_2Te_4 . Table 5 therefore records only its two source-specific $J_\perp$ – K sets instead of filling the missing quantities from another material or model.

3 Spin Hamiltonians

3.1 Atomistic Spin Hamiltonian

Equation. Following the ASTRO exchange-sign convention, a classical spin Hamiltonian containing exchange, uniaxial anisotropy, Dzyaloshinskii–Moriya interaction, and Zeeman energy is

$$\begin{aligned} \mathcal{H} = & + \frac{1}{2} \sum_{i \neq j} J_{ij} \mathbf{s}_i \cdot \mathbf{s}_j - \sum_i K_i (\mathbf{s}_i \cdot \hat{\mathbf{e}}_i)^2 \\ & - \frac{1}{2} \sum_{i \neq j} \mathbf{D}_{ij} \cdot (\mathbf{s}_i \times \mathbf{s}_j) - \sum_i \boldsymbol{\mu}_i \cdot \mathbf{B}_{\text{ext}}. \end{aligned} \quad (3.1)$$

Definitions. J_{ij} is the exchange-link energy in the ASTRO sign convention: $J_{ij} < 0$ favors parallel alignment, whereas $J_{ij} > 0$ favors antiparallel alignment. K_i is the uniaxial anisotropy energy, $\hat{\mathbf{e}}_i$ is the easy-axis direction, $\mathbf{D}_{ij}$ is the Dzyaloshinskii–Moriya vector, and $\boldsymbol{\mu}_i$ is the magnetic moment at site i .

Assumptions. Each $\mathbf{s}_i$ is a classical unit vector. The pair coefficients satisfy $J_{ji} = J_{ij}$ and $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$. The factors of 1/2 assume that the pair sums include both ordered pairs (i, j) and (j, i) . If each physical bond is instead included only once, for example in a unique-bond sum $\sum_{\langle i, j \rangle}$, both factors of 1/2 must be omitted. These are equivalent pair-counting conventions and must not be mixed.

Layered exchange example. For ferromagnetic exchange within a plane and antiferromagnetic exchange between planes, the ASTRO sign convention in Eq. (3.1) gives [11]

$$J_\parallel < 0, \quad J_\perp > 0, \quad (3.2)$$

where $J_\parallel$ couples in-plane neighbors and $J_\perp$ couples neighbors in adjacent planes. A source that puts a minus sign in front of the exchange sum uses the opposite signs for $J_\parallel$ and $J_\perp$. The corresponding A-type arrangement is described in *Spintronics Concepts and Symmetry*; its structural classification remains **Reference needed**.

Figure 1 collects the geometric quantities attached to one bond and distinguishes a physical bond from its two ordered appearances in the Hamiltonian sum.

Pair counting and effective field. Let $\mu_i = |\boldsymbol{\mu}_i|$ and define the effective flux density by

$$\mathbf{B}_{\text{eff},i} = -\frac{1}{\mu_i} \frac{\partial \mathcal{H}}{\partial \mathbf{s}_i}. \quad (3.3)$$

For the ordered-pair Hamiltonian above, the exchange and DMI contributions are

$$\mathbf{B}_{\text{ex},i} = -\frac{1}{\mu_i} \sum_{j \neq i} J_{ij} \mathbf{s}_j, \quad \mathbf{B}_{\text{DMI},i} = \frac{1}{\mu_i} \sum_{j \neq i} \mathbf{s}_j \times \mathbf{D}_{ij}. \quad (3.4)$$

The two ordered occurrences of each bond give identical derivatives, which cancel the Hamiltonian factors of 1/2. Consequently, no extra factor of 1/2 belongs in either effective-field expression.

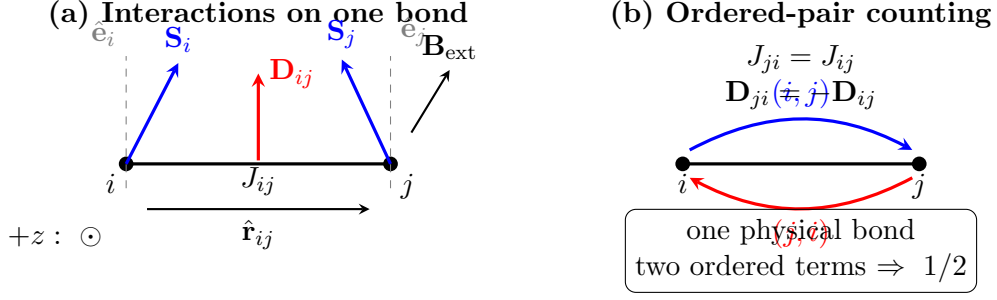


Figure 1: Original schematic of the bond quantities in Eq. (3.1). Panel (a) shows two spins, their local easy axes, the exchange link, the directed bond, the DMI vector, and the external field. The displayed DMI orientation illustrates the ASTRO convention $\mathbf{D}_{ij} = D(\hat{\mathbf{z}} \times \hat{\mathbf{r}}_{ij})$. Panel (b) shows why an ordered-pair sum contains both (i, j) and (j, i) and therefore carries a factor $1/2$ [11].

Units. All terms in $\mathcal{H}$ have units of energy. Here the external flux density $\mathbf{B}_{\text{ext}}$ is measured in tesla, so no additional factor of μ_0 appears in the Zeeman term.

Source. For the extended classical spin Hamiltonian and Hamiltonian-derivative field convention, see Evans et al. [2].

Implementation notes. The ASTRO implementation evaluates the field, $\mathbf{B}_{\text{eff},i} = -(1/\mu_i) \partial \mathcal{H} / \partial \mathbf{s}_i$, rather than the total energy and includes both forward and backward nearest neighbors. The J_{ij} used above is therefore the coefficient stored in ASTRO’s exchange-link arrays: its exchange field is $-(1/\mu_i) \sum_j J_{ij} \mathbf{s}_j$ [11]. No factor of $1/2$ should appear in the exchange or DMI field formulas because differentiation produces two identical bond contributions, cancelling the Hamiltonian factor. ASTRO’s DMI component expressions correspond to $\mathbf{D}_{ij} = D(\hat{\mathbf{z}} \times \hat{\mathbf{r}}_{ij})$ and the antisymmetry $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$ used above.

3.2 Boundary Conditions in Atomistic Model Development

Boundary conditions are part of the physical model, but the most useful development order need not be the final production geometry. A practical atomistic workflow is to establish the bulk model under periodic boundary conditions first and then change to open boundaries when the scientific question concerns a finite sample, an edge, or a spatially nonuniform process. This is a verification sequence, not a rule that periodic boundaries are always more physical [11, 3].

Periodic bulk-verification stage. Periodic boundaries identify opposite sides of the simulation cell, so a displayed edge is connected to its periodic image and is not a physical surface. This makes periodic calculations useful for learning and checking the lattice construction, bulk coordination, bond multiplicities, pair counting, translational invariance, energy per cell, and supercell invariance. Small deterministic lattices should also verify the orientation of every wrapped directed bond. The ASTRO reference implementation and its field tests treat periodic and nonperiodic neighbor semantics as distinct cases rather than changing them implicitly [11].

Open finite-geometry stage. After the structure, interaction topology, conservative fields, dynamical right-hand side, and integrator have passed their periodic checks, use open boundaries for spatially resolved finite-device studies. In an atomistic open-boundary model, edge spins remain dynamical and a finite-range bond is omitted when its other endpoint lies outside the simulated geometry. This must not be implemented by wrapping the neighbor to the opposite edge, clamping the edge spin, or inserting a zero-valued ghost spin. Fixed or pinned spins

describe an additional physical constraint and require their own justification. Mixed boundaries are appropriate when the target geometry is periodic along one direction and finite along another [11].

DMI and physical edge response. For the unique-bond DMI convention in Eq. (3.1), every retained directed bond must obey $\mathbf{D}_{ji} = -\mathbf{D}_{ij}$. An open edge omits DMI bonds that cross the boundary but does not suppress the response of the remaining edge spins. The reduced edge coordination can produce chiral canting, boundary-localized textures, and edge-assisted nucleation; such effects are part of the finite magnetic problem rather than automatically numerical artifacts. Spatially resolved atomistic SOT calculations with DMI provide an example in which local switching behavior cannot be inferred safely from only a spatial average [3]. Truncating bulk DMI bonds is the simplest free-edge approximation; it does not establish that a real atomic termination has unchanged bulk bond coefficients.

Finite-size and wave checks. Open boundaries remove direct periodic wraparound, but two edges can still interact through a domain wall, soliton, or propagating spin wave when the sample is too small. Compare increasing lateral sizes and require both the central-region trajectory and the selected device observable to converge, while retaining edge-resolved diagnostics. Periodic calculations require a corresponding check that a localized texture does not interact materially with its periodic images. Neither an ordinary open boundary nor a periodic boundary is an absorbing boundary. If suppression of reflected waves is a numerical requirement, introduce and validate a separate damping buffer and report the observable region from which that buffer is excluded [15, 11].

The recommended sequence is therefore

$$\text{periodic bulk construction and term checks} \longrightarrow \text{open finite-geometry spatial studies}, \quad (3.5)$$

with both modes retained as explicit, tested configurations. A result must state which mode was used; changing the boundary mode changes the modeled system and is not merely a solver option [11, 3].

3.3 Macrospin Effective Fields

Field convention. For a fixed-length macrospin with normalized magnetization $\mathbf{m}$, let w denote the magnetic energy density in J m^{-3} . The effective magnetic field and the corresponding effective flux density are [34]

$$\mathbf{H}_{\text{eff}} = -\frac{1}{\mu_0 M_s} \frac{\partial w}{\partial \mathbf{m}}, \quad \mathbf{B}_{\text{eff}} = \mu_0 \mathbf{H}_{\text{eff}}. \quad (3.6)$$

Thus, an LLG equation written with $\gamma_0 = \mu_0 |\gamma|$ uses $\mathbf{H}_{\text{eff}}$ in A m^{-1} , whereas one written with $|\gamma|$ uses $\mathbf{B}_{\text{eff}}$ in tesla. The two conventions are physically equivalent but their field variables must not be mixed.

Unless stated otherwise, K_u and A_{12} are energy densities in J m^{-3} . The interfacial K_i in the combination $K_{\text{bulk}} + K_i/t_{\text{FM}}$ is instead in J m^{-2} , and every saturation magnetization is in A m^{-1} . The tensors $\mathbf{N}$ and $\mathbf{K}_{fp}$ are dimensionless.

3.3.1 One-Sublattice Model

Energy density and field decomposition. For uniaxial anisotropy with easy axis $\hat{\mathbf{u}}$, a dimensionless symmetric demagnetization tensor $\mathbf{N}$, an applied field $\mathbf{H}_{\text{ext}}$, and a dipolar field $\mathbf{H}_{\text{dip}}$ produced by another magnetic body, use the uniaxial-anisotropy convention from slides 13–14 of the EE237 Week 5 lecture notes [34] and take

$$w_1 = K_u \left[1 - (\mathbf{m} \cdot \hat{\mathbf{u}})^2 \right] + \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} - \mu_0 M_s \mathbf{m} \cdot (\mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{dip}}). \quad (3.7)$$

External field. The part of the final term in Eq. (3.7) that is proportional to $\mathbf{H}_{\text{ext}}$ is the Zeeman coupling to the applied field. Its derivative contributes $\mathbf{H}_{\text{ext}}$ directly to $\mathbf{H}_{\text{eff}}$ [34].

Uniaxial anisotropy field. Equation (3.6) gives [34]

$$\mathbf{H}_{\text{an}} = \frac{2K_u}{\mu_0 M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}, \quad \mathbf{B}_{\text{an}} = \frac{2K_u}{M_s} (\mathbf{m} \cdot \hat{\mathbf{u}}) \hat{\mathbf{u}}. \quad (3.8)$$

For $\hat{\mathbf{u}} = \hat{\mathbf{z}}$, the anisotropy term reduces to $K_u(1 - m_z^2) = K_u \sin^2 \theta$, as on slide 13. Near $m_z = 1$, slide 14 therefore gives $H_{\text{an},z} = 2K_u/(\mu_0 M_s)$. For a thin film with bulk and interfacial uniaxial anisotropy, $K_u = K_{\text{bulk}} + K_i/t_{\text{FM}}$.

Figure 2 connects the magnetization geometry to the zero-field uniaxial energy barrier used in the thermal-stability factor.

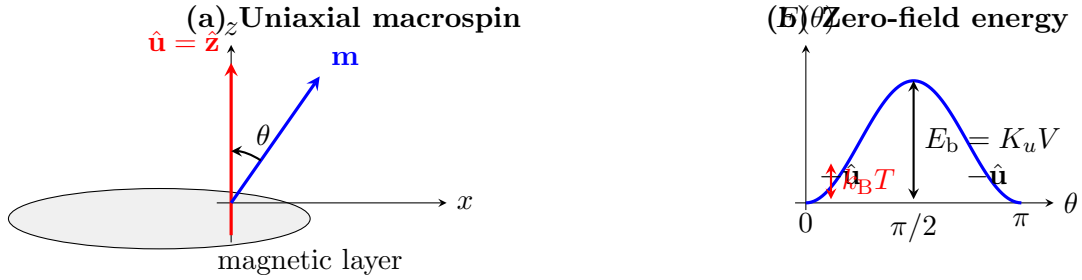


Figure 2: Original schematic linking uniaxial macrospin geometry and thermal stability. (a) The normalized magnetization $\mathbf{m}$ makes an angle θ with the easy axis $\hat{\mathbf{u}}$. (b) For a zero-field volume V with energy $E(\theta) = K_u V \sin^2 \theta$, the two easy-axis states are separated by $E_b = K_u V$; the ratio of this barrier to the thermal scale $k_B T$ is the stability factor. The conventions follow the EE237 anisotropy and thermal-stability notes [34, 35].

Demagnetization field. Slide 21 of the same notes writes the demagnetization self-energy as [34]

$$E_{\text{demag}} = -\frac{\mu_0}{2} \int_V \mathbf{M}(\mathbf{r}) \cdot \mathbf{H}_{\text{demag}}(\mathbf{r}) d^3 r, \quad (3.9)$$

$$w_{\text{demag}} = -\frac{\mu_0}{2} \mathbf{M} \cdot \mathbf{H}_{\text{demag}} = \frac{\mu_0 M_s^2}{2} \mathbf{m} \cdot \mathbf{N} \mathbf{m} \quad (\text{uniform macrospin}). \quad (3.10)$$

The factor 1/2 avoids double counting the magnet's self-interaction. Equation (3.6) gives

$$\mathbf{H}_{\text{demag}} = -M_s \mathbf{N} \mathbf{m}, \quad \mathbf{B}_{\text{demag}} = -\mu_0 M_s \mathbf{N} \mathbf{m}. \quad (3.11)$$

For principal axes, $\mathbf{N} = \text{diag}(N_x, N_y, N_z)$ and $N_x + N_y + N_z = 1$.

Dipolar field. Let a pinned macrospin have volume V_p , saturation magnetization $M_{s,p}$, and direction $\mathbf{m}_p$, so its magnetic moment is $\boldsymbol{\mu}_p = V_p M_{s,p} \mathbf{m}_p$. In the point-dipole approximation, at displacement $\mathbf{r} = r \hat{\mathbf{r}}$ from the pinned layer to the free layer, slide 22 gives the dipole formula [34]. The slide denotes it by $\mathbf{H}_i$, but its factor μ_0 makes it a flux density in tesla; therefore, in the convention of this document,

$$\mathbf{B}_{\text{dip}} = \frac{\mu_0}{4\pi r^3} [3(\boldsymbol{\mu}_p \cdot \hat{\mathbf{r}}) \hat{\mathbf{r}} - \boldsymbol{\mu}_p] = \mu_0 \mathbf{H}_{\text{dip}}, \quad (3.12)$$

Thermal field. Following Eq. (4) of Zhu et al. [12], write the sampled thermal field as

$$\mathbf{B}_{\text{th}} = \mu_0 \mathbf{H}_{\text{th}} = \mathcal{N}_1(0, u_B) \hat{\mathbf{x}} + \mathcal{N}_2(0, u_B) \hat{\mathbf{y}} + \mathcal{N}_3(0, u_B) \hat{\mathbf{z}}, \quad (3.13)$$

$$u_B = \sqrt{\frac{2\alpha k_B T}{V_{\text{FL}} M_s |\gamma| (1 + \alpha^2) \Delta t}}. \quad (3.14)$$

Here the $\mathcal{N}_i(0, u_B)$ are independent normal random variables with zero mean and standard deviation u_B , V_{FL} is the free-layer volume, and Δt is the duration over which a sampled field is held constant. With $|\gamma|$ in $\text{rad s}^{-1} \text{T}^{-1}$ and M_s in A m^{-1} , u_B is in tesla; the corresponding $\mathbf{H}$ -field standard deviation is $u_H = u_B / \mu_0$.

The source uses independent samples over $\Delta t = 5 \text{ ps}$ [12]. When adopting another integration interval or LLG convention, recompute the noise amplitude consistently rather than reusing that numerical value.

Total field. The total one-sublattice field, with the stochastic field added after taking the energy derivative, is [33]

$$\mathbf{H}_{\text{eff}} = \mathbf{H}_{\text{ext}} + \mathbf{H}_{\text{an}} + \mathbf{H}_{\text{demag}} + \mathbf{H}_{\text{dip}} + \mathbf{H}_{\text{th}}. \quad (3.15)$$

Spin-transfer and spin-orbit torques are nonconservative and are not part of the energy derivative in Eq. (3.6).

Implementation mapping. The current one-sublattice `ODE_LLG_integral` implementation stores flux-density quantities in tesla [33]. Its `field_eta.m` evaluates the following expressions, where its input named `Hk` is B_k :

$$\begin{aligned} \mathbf{B}_{\text{an}} &= B_k m_y \hat{\mathbf{y}} \quad (\text{IMA}), & \mathbf{B}_{\text{an}} &= B_k m_z \hat{\mathbf{z}} \quad (\text{PMA}), \\ \mathbf{B}_{\text{demag}} &= -\mu_0 M_s \mathbf{N} \mathbf{m}, & \mathbf{B}_{\text{dip}} &= \mu_0 \mathbf{K}_{12} M_s \mathbf{m}_{\text{PL}}, \end{aligned}$$

after converting the input M_s from emu cm^{-3} to A m^{-1} . It sums these with the applied and thermal fields.

3.3.2 Two-Sublattice Collinear Model

Inter-sublattice exchange field. Let $\mathbf{m}_1$ and $\mathbf{m}_2$ be separately normalized sublattice magnetizations with saturation magnetizations $M_{s,1}$ and $M_{s,2}$. The exchange-energy density convention used by the two-sublattice ferrimagnetic macrospin model of Yuan et al. [13] is

$$w_{\text{ex},12} = A_{12} \mathbf{m}_1 \cdot \mathbf{m}_2, \quad A_{12} > 0, \quad (3.16)$$

Applying $\mathbf{H}_{\text{ex},i} = -(\mu_0 M_{s,i})^{-1} \partial w_{\text{ex},12} / \partial \mathbf{m}_i$ yields, for $j \neq i$,

$$\mathbf{H}_{\text{ex},i} = -\frac{A_{12}}{\mu_0 M_{s,i}} \mathbf{m}_j, \quad \mathbf{B}_{\text{ex},i} = -\frac{A_{12}}{M_{s,i}} \mathbf{m}_j. \quad (3.17)$$

Only the exchange contribution unique to the two-sublattice model is stated here; the shared external, anisotropy, demagnetization, dipolar, and thermal field formulas are defined above and are not repeated. This two-sublattice exchange extension is publication-grounded and is not implemented by the `ODE_LLG_integral` repository.

4 Magnetization Dynamics

LLGS form with field-like torque. This macrospin equation is adapted from Eqs. (1)–(4) of Kong *et al.*, with the atom index removed [3]. It is written in the flux-density convention: $\mathbf{B}_{\text{eff}}$ and the torque amplitudes are in tesla and are paired with γ in $\text{rad s}^{-1} \text{T}^{-1}$.

$$\begin{aligned} \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{B}_{\text{eff}} + \alpha \mathbf{m} \times \frac{d\mathbf{m}}{dt} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (4.1)$$

Figure 3 fixes the device axes and illustrates the vector structures that appear in Eq. (4.1).

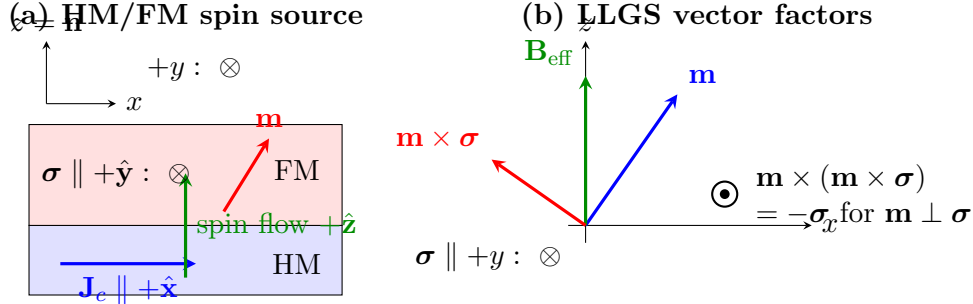


Figure 3: Original schematic of a conventional spin-Hall-driven LLGS geometry. (a) Charge current along $+x$ in the heavy metal generates a spin flow along the interface normal $+z$ with the illustrated polarization along $+y$; reversing the current or the signed spin-Hall angle reverses the polarization. (b) For $\mathbf{m}$ in the xz plane, the field-like vector $\mathbf{m} \times \boldsymbol{\sigma}$ lies in that plane, whereas the damping-like vector $\mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma})$ points along $-y$. These are vector factors only; their signed scalar prefactors are those in Eq. (4.1). The conventions follow the SOT compact model and Kong *et al.* [5, 3].

Equivalent LL form with field-like torque. The algebraically equivalent form follows the same source convention [3].

$$\begin{aligned} (1 + \alpha^2) \frac{d\mathbf{m}}{dt} = & -\gamma \mathbf{m} \times \mathbf{B}_{\text{eff}} - \alpha \gamma \mathbf{m} \times (\mathbf{m} \times \mathbf{B}_{\text{eff}}) \\ & + \alpha \gamma \tau_{\text{DLT}} \mathbf{m} \times \boldsymbol{\sigma} - \gamma \tau_{\text{DLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) \\ & + \alpha \gamma \tau_{\text{FLT}} \mathbf{m} \times (\mathbf{m} \times \boldsymbol{\sigma}) + \gamma \tau_{\text{FLT}} \mathbf{m} \times \boldsymbol{\sigma}. \end{aligned} \quad (4.2)$$

Torque amplitudes. For a ferromagnetic layer [4],

$$\tau_{\text{DLT}} = \frac{\hbar \theta_{\text{SH}} J_c}{2e t_{\text{FM}} M_s}, \quad \tau_{\text{FLT}} = \beta \tau_{\text{DLT}}.$$

Torque-sign convention. In the convention of Eq. (4.1), the sign product of the damping-like and field-like SOT is negative when $\beta > 0$ [1, 3]. Specifically, the scalar prefactors of their respective vector structures are $-\gamma \tau_{\text{DLT}}$ and $+\gamma \tau_{\text{FLT}}$, so their sign product is

$$(-\gamma \tau_{\text{DLT}})(+\gamma \tau_{\text{FLT}}) = -\gamma^2 \beta \tau_{\text{DLT}}^2 < 0 \quad (\beta > 0).$$

Thus, a positive β means that the two SOT terms have oppositely signed scalar prefactors in the LLGS equation, even though the signed amplitudes τ_{DLT} and τ_{FLT} themselves have the same sign.

5 Thermal Stability

5.1 Thermal Stability Factor

Equation. For a single energy barrier E_b , the thermal stability factor is [35]

$$\Delta = \frac{K_u V}{k_B T} \quad (5.1)$$

The zero-field uniaxial example in Fig. 2 shows the barrier $E_b = K_u V$ relative to the thermal energy scale $k_B T$.

Units. E_b and $k_B T$ must use the same energy unit; Δ is dimensionless.

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- [35] EE237 Week 10 lecture slides, MRAM Cells, Reliability, and Design Tradeoffs, slide 28.
- [36] EE237 Week 11 lecture slides, Spin–Orbit Torques and Switching Symmetry, slides 4–5.